Explicit construction for $J_1 = J_2 = \frac{1}{2}$; $J_1 = 1$, $J_2 = \frac{1}{2}$; $J_1 = \ell$, $J_2 = \frac{1}{2}$.

① $J_1 = \frac{1}{2}$, $J_2 = \frac{1}{2}$ $\Rightarrow$ $0 = \frac{1}{2} - \frac{1}{2} \leq J \leq \frac{1}{2} + \frac{1}{2} = 1$

$\Rightarrow$ 4 states: 3 with $J = 1$ $(m = -1, 0, 1)$ and 1 with $J = 0$ $(m = 0)$.

<u>Original</u> states are $|\frac{1}{2}, \pm\frac{1}{2}; \frac{1}{2}, \pm\frac{1}{2}\rangle \xrightarrow[\text{simplify notation}]{} |\psi_{\pm\frac{1}{2}, \pm\frac{1}{2}}\rangle$

<u>New states</u> are eigenstates of total $\vec{J}^2$ and $J_z$ (as well as $\vec{J}_1^2, \vec{J}_2^2$)

Denote as $|\phi_{j,m}\rangle$ for brevity.

Clearly $|\phi_{1,1}\rangle = |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle$ as there is only 1 state with $m = 1 = \frac{1}{2} + \frac{1}{2} = m_1 + m_2$.

To find $|\phi_{1,0}\rangle$ we act with $\hat{J}_-$ on $|\phi_{1,1}\rangle$:

By def $\hat{J}_- |\phi_{1,1}\rangle = \sqrt{j(j+1) - m(m-1)}\, |\phi_{1,\underbrace{1-1}_{0}}\rangle = \sqrt{2}\, |\phi_{1,0}\rangle$

But in the old basis

$$\hat{J}_- |\phi_{1,1}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) |\psi_{\frac{1}{2}, \frac{1}{2}}\rangle = |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}} \left( |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle + |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle \right).$$

Then $\hat{J}_- |\phi_{1,0}\rangle = \sqrt{2 - 0}\, |\phi_{1,-1}\rangle = \sqrt{2}\, |\phi_{1,-1}\rangle$

but $\hat{J}_- |\phi_{1,0}\rangle = (\hat{J}_{1-} + \hat{J}_{2-}) \frac{1}{\sqrt{2}} \left( |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle + |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right) =$

$$= \frac{1}{\sqrt{2}} 2 |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle = \sqrt{2}\, |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle$$

$$\Rightarrow |\phi_{1,-1}\rangle = |\psi_{-\frac{1}{2}, -\frac{1}{2}}\rangle.$$

The remaining state $|\phi_{0,0}\rangle$ must be orthogonal to $|\phi_{1,0}\rangle$

$$\Rightarrow |\phi_{0,0}\rangle = \frac{1}{\sqrt{2}} \left( |\psi_{\frac{1}{2}, -\frac{1}{2}}\rangle - |\psi_{-\frac{1}{2}, \frac{1}{2}}\rangle \right).$$ <u>Singlet</u>

by inspection

Naturally $|\phi_{1,m}\rangle$ = triplet states.

(2) Spin $1/2$ + Orbital angular momentum $l$

$\vec{J} = \vec{L} + \vec{S}$ . $\vec{S}^2 = \frac{3}{4}\hbar^2$

Old states are $|l, m_l; s, m_s\rangle \Rightarrow |\psi_{m_l m_s}\rangle$

New states $|j, m; l, s\rangle \Rightarrow |\phi_{jm}\rangle$ , $m = m_l + m_s$

Two possibilities: $\begin{cases} j = l + \frac{1}{2} & \text{ladder contains } 2j+1 = \underline{2l+2} \text{ states} \\ j = l - \frac{1}{2} & \text{ladder contains } 2j+1 = 2l-1+1 = \underline{2l} \end{cases}$

$\Rightarrow$ total # states $2l+2+2l = 2(2l+1)$ which is exactly the dimension of $|\psi\rangle$ space.

Start with $\underline{\underline{j = l + \frac{1}{2}}}$ : $|\phi_{l+\frac{1}{2}, m}\rangle = \sum_{m_l + m_s = m} c_{m_l m_s} |\psi_{m_l, m_s}\rangle =$

$$= \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle$$

$m_l = m - \frac{1}{2}$, $m_s = \frac{1}{2}$ $\qquad$ $m_l = m + \frac{1}{2}$, $m_s = -\frac{1}{2}$

Normalization: $|\alpha_m|^2 + |\beta_m|^2 = 1$

Apply lowering operator $\hat{J}_- |\phi_{l+\frac{1}{2}, m}\rangle \overset{\text{by def}}{=} \hbar\sqrt{j(j+1) - m(m-1)}\, |\phi_{l+\frac{1}{2}, m-1}\rangle$

(here $j = l + \frac{1}{2}$)

$$= \hbar\sqrt{(j+m)(j-m+1)}\, |\phi_{l+\frac{1}{2}, m-1}\rangle = \hbar\sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{3}{2})} \cdot$$

($j = l + \frac{1}{2}$)

$$\cdot \left\{ \alpha_{m-1} |\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \beta_{m-1} |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right\} . \quad (1)$$

On the other hand

$$\hat{J}_- |\phi_{l+\frac{1}{2}, m}\rangle = (\hat{L}_- + \hat{S}_-) \left\{ \alpha_m |\psi_{m-\frac{1}{2}, \frac{1}{2}}\rangle + \beta_m |\psi_{m+\frac{1}{2}, -\frac{1}{2}}\rangle \right\}$$

$$= \hbar\, \alpha_m \sqrt{l(l+1) - (m-\tfrac{1}{2})(m-\tfrac{3}{2})} \left( |\psi_{m-\frac{3}{2}, \frac{1}{2}}\rangle + \sqrt{\tfrac{1}{2}(1+\tfrac{1}{2}) - \tfrac{1}{2}(\tfrac{1}{2}-1)}\, |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle \right)$$

$$+ \hbar\, \beta_m \sqrt{l(l+1) - (m+\tfrac{1}{2})(m-\tfrac{1}{2})} \left( |\psi_{m-\frac{1}{2}, -\frac{1}{2}}\rangle + \underbrace{\sqrt{\tfrac{1}{2}(\tfrac{1}{2}+1) - (-\tfrac{1}{2})(-\tfrac{1}{2}-1)}}_{\emptyset}\, \underbrace{|\psi_{m+\frac{1}{2}, -\frac{1}{2}-1}\rangle}_{\text{does not exist}} \right)$$

Use $l(l+1) - m(m-1) = (l+m)(l-m+1)$.

Thus $\hat{J}_- |\phi_{l+\frac{1}{2},m}\rangle = \hbar\alpha_m \sqrt{(l+m-\frac{1}{2})(l-m+\frac{3}{2})}\, |\psi_{m-\frac{3}{2},\frac{1}{2}}\rangle +$

$+ |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle) + \hbar\, \beta_m \sqrt{(l+m+\frac{1}{2})(l-m+\frac{1}{2})}\, |\psi_{m-\frac{1}{2},-\frac{1}{2}}\rangle$. (2)

Right-hand sides of (1) and (2) are the same $\Rightarrow$ we can compare coefficients:

$$\alpha_{m-1}\sqrt{(l+m+\tfrac{1}{2})(l-m+\tfrac{3}{2})} = \alpha_m \sqrt{(l+m-\tfrac{1}{2})(l-m+\tfrac{3}{2})} \quad (3)$$

(The relation for $\beta_m$ obtains from normalization: $|\beta_m|^2 = 1 - |\alpha_m|^2$.)

Observe that (3):

$$\frac{\alpha_m}{\sqrt{l+m+\frac{1}{2}}} = \frac{\alpha_{m-1}}{\sqrt{l+m-\frac{1}{2}}} = r$$ ($r$ is constant independent of $m$!)

Now, for a maximum possible $m = l+\frac{1}{2}$:

$$|\phi_{l+\frac{1}{2},l+\frac{1}{2}}\rangle = \alpha_{l+\frac{1}{2}} |\psi_{l,\frac{1}{2}}\rangle + \beta_{l+\frac{1}{2}} |\psi_{l+1,-\frac{1}{2}}\rangle$$

state with $m_\ell = l+1$, $m_s = -\frac{1}{2}$.

But $m_\ell \le l$ always

$\Rightarrow \boxed{\beta_{l+\frac{1}{2}} = 0}$

$\Rightarrow \alpha_{l+\frac{1}{2}} = 1$.

$$\Rightarrow r = \frac{\alpha_{l+\frac{1}{2}}}{\sqrt{l+l+\frac{1}{2}+\frac{1}{2}}} = \frac{1}{\sqrt{2l+1}}$$

$$\Rightarrow \alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$$

$$|\beta_m|^2 = 1 - \frac{l+m+\frac{1}{2}}{2l+1} = \frac{l-m+\frac{1}{2}}{2l+1} \Rightarrow \beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$$

Thus $|\phi_{l+\frac{1}{2},m}\rangle = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle + \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$

The states from the other ladder, with $\underline{\underline{j=l-\frac{1}{2}}}$;

must be orthogonal to $|\phi_{l+\frac{1}{2},m}\rangle$ at the same $\underline{m}$:

Thus

$$|\phi_{l-\frac{1}{2},m}\rangle = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle - \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

Clearly $\langle\phi_{l+\frac{1}{2},m}|\phi_{l-\frac{1}{2},m}\rangle = 0$ always.

Thus we found Clebsch-Gordan coefficients for $\vec{J} = \vec{L} + \vec{S}$, $S = \frac{1}{2}$.

They are $\alpha_m = \sqrt{\frac{l+m+\frac{1}{2}}{2l+1}}$ and $\beta_m = \sqrt{\frac{l-m+\frac{1}{2}}{2l+1}}$.

<u>The limit $l=\frac{1}{2}$:</u> $\alpha_m = \sqrt{\frac{1+m}{2}}$ $\quad \beta_m = \sqrt{\frac{1-m}{2}}$

$$|\phi_{l=1,m}\rangle = \sqrt{\frac{1+m}{2}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle + \sqrt{\frac{1-m}{2}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle$$

$$|\phi_{0,m}\rangle = \sqrt{\frac{1-m}{2}}\,|\psi_{m-\frac{1}{2},\frac{1}{2}}\rangle - \sqrt{\frac{1+m}{2}}\,|\psi_{m+\frac{1}{2},-\frac{1}{2}}\rangle =$$

$$= (\text{but } m=0 \text{ the only option}) = \frac{1}{\sqrt{2}}\left(|-\tfrac{1}{2},\tfrac{1}{2}\rangle - |\tfrac{1}{2},-\tfrac{1}{2}\rangle\right)$$

Singlet!

<u>triplet</u>
$m=+1,0,-1$

$$\begin{cases} |\phi_{1,+1}\rangle = |\frac{1}{2},\frac{1}{2}\rangle \\ |\phi_{1,0}\rangle = \frac{1}{\sqrt{2}}\left(|-\frac{1}{2},\frac{1}{2}\rangle + |\frac{1}{2},-\frac{1}{2}\rangle\right) \\ |\phi_{1,-1}\rangle = |-\frac{1}{2},-\frac{1}{2}\rangle \end{cases}$$